

## SAFETY DATA SHEET

Creation Date 02-February-2010

Revision Date 25-March-2024

Revision Number 2

### 1. Identification

**Product Name** Zinc sulfate heptahydrate

**Cat No. :** S60069

**CAS-No** 7446-20-0  
**Synonyms** zinc vitriol.; White vitriol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |            |
|-------------------------------------------------------------|------------|
| <b>Acute oral toxicity</b>                                  | Category 4 |
| <b>Serious Eye Damage/Eye Irritation</b>                    | Category 1 |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 2 |
| Target Organs - Heart, Blood.                               |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Harmful if swallowed

Causes serious eye damage

May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                 | CAS-No    | Weight % |
|---------------------------|-----------|----------|
| Zinc sulfate heptahydrate | 7446-20-0 | 100      |
| Zinc sulfate              | 7733-02-0 | -        |

### 4. First-aid measures

|                                                               |                                                                                                                                   |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.           |
| <b>Inhalation</b>                                             | Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention immediately if symptoms occur.                 |
| <b>Ingestion</b>                                              | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                    |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | Causes eye burns.<br>Treat symptomatically                                                                                        |

### 5. Fire-fighting measures

|                                                      |                                                                                       |
|------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                  | Substance is nonflammable; use agent most appropriate to extinguish surrounding fire. |
| <b>Unsuitable Extinguishing Media</b>                | No information available                                                              |
| <b>Flash Point<br/>Method -</b>                      | Not applicable<br>No information available                                            |
| <b>Autoignition Temperature<br/>Explosion Limits</b> | No information available                                                              |

|                                  |                          |
|----------------------------------|--------------------------|
| Upper                            | No data available        |
| Lower                            | No data available        |
| Sensitivity to Mechanical Impact | No information available |
| Sensitivity to Static Discharge  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Sulfur oxides.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

Health  
2

Flammability  
0

Instability  
0

Physical hazards  
N/A

**6. Accidental release measures****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Avoid dust formation. Sweep up and shovel into suitable containers for disposal.

**7. Handling and storage****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Do not breathe dust. Avoid contact with skin, eyes or clothing.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong bases.

**8. Exposure controls / personal protection****Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Natural rubber | See manufacturers recommendations |                 | Splash protection only |
| Nitrile rubber |                                   |                 |                        |

Neoprene  
PVC

-

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Solid                    |
| Appearance                             | White                    |
| Odor                                   | Odorless                 |
| Odor Threshold                         | No information available |
| pH                                     | 4.4-6 5% aq. solution    |
| Melting Point/Range                    | 100 °C / 212 °F          |
| Boiling Point/Range                    | No information available |
| Flash Point                            | Not applicable           |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 3.54 @ 25°C              |
| Solubility                             | Soluble in water         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | 500°C                    |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | O4 S Zn . 7 H2 O         |
| Molecular Weight                       | 287.53                   |

## 10. Stability and reactivity

|                        |                                                           |
|------------------------|-----------------------------------------------------------|
| Reactive Hazard        | None known, based on information available                |
| Stability              | Stable under normal conditions.                           |
| Conditions to Avoid    | Avoid dust formation. Incompatible products. Excess heat. |
| Incompatible Materials | Strong bases                                              |

**Hazardous Decomposition Products** Sulfur oxides

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component                 | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation |
|---------------------------|---------------------------|---------------------------|-----------------|
| Zinc sulfate heptahydrate | 1260 mg/kg ( Rat )        | Not listed                | Not listed      |
| Zinc sulfate              | LD50 = 1710 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | Not listed      |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** Severe eye irritant

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                 | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------|-----------|------------|------------|------------|------------|------------|
| Zinc sulfate heptahydrate | 7446-20-0 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Zinc sulfate              | 7733-02-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** Experiments have shown reproductive toxicity effects on laboratory animals.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** Heart Blood

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** Tumorigenic effects have been reported in experimental animals. See actual entry in RTECS for complete information.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                 | Freshwater Algae                                     | Freshwater Fish                                          | Microtox                                           | Water Flea                                           |
|---------------------------|------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------|------------------------------------------------------|
| Zinc sulfate heptahydrate | Not listed                                           | 1.9 mg/L LC50 96 h                                       | Not listed                                         | Not listed                                           |
| Zinc sulfate              | EC50: = 0.056 mg/L, 72h static (Pseudokirchneriella) | LC50: 0.48 - 1.72 mg/L, 96h static (Poecilia reticulata) | EC50 = 3.45 mg/L 15 min<br>EC50 = 40.5 mg/L 30 min | EC50: 0.538 - 0.908 mg/L, 48h Static (Daphnia magna) |

|  |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                               |                                        |
|--|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------|
|  | subcapitata) | LC50: 49.23 - 64.16 mg/L, 96h semi-static (Poecilia reticulata)<br>LC50: = 0.63 mg/L, 96h (Poecilia reticulata)<br>LC50: 3.55 - 6.32 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 3 - 4.6 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: 16.85 - 27.18 mg/L, 96h static (Cyprinus carpio)<br>LC50: = 0.162 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.168 - 0.25 mg/L, 96h semi-static (Pimephales promelas)<br>LC50: 0.23 - 0.48 mg/L, 96h (Pimephales promelas)<br>LC50: = 0.06 mg/L, 96h static (Pimephales promelas)<br>LC50: 0.218 - 0.42 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 0.34 - 0.93 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 0.03 - 0.05 mg/L, 96h semi-static (Oncorhynchus mykiss)<br>LC50: = 0.15 mg/L, 96h semi-static (Cyprinus carpio) | EC50 = 476 mg/L 5 min<br>EC50 > 700 mg/L 16 h | EC50: = 0.75 mg/L, 48h (Daphnia magna) |
|--|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------|

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3077  
 Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
 Hazard Class 9  
 Packing Group III

#### TDG

UN-No UN3077  
 Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
 Hazard Class 9  
 Packing Group III

#### IATA

UN-No UN3077  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.\*  
 Hazard Class 9  
 Packing Group III

#### IMDG/IMO

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

## 15. Regulatory information

### International Inventories

| Component                 | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Zinc sulfate heptahydrate | 7446-20-0 | X   | -    | -    | -                                             | -         | -      | -   |
| Zinc sulfate              | 7733-02-0 | X   | -    | X    | ACTIVE                                        | 231-793-3 | -      | -   |

| Component                 | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Zinc sulfate heptahydrate | 7446-20-0 | X     | -        | X    | -    | X    | X    | X     | X     |
| Zinc sulfate              | 7733-02-0 | X     | KE-35582 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                 | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|---------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Zinc sulfate heptahydrate | Part 1, Group A Substance                            |                                                                            |                                           |
| Zinc sulfate              | Part 1, Group A Substance                            |                                                                            |                                           |

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component                 | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Zinc sulfate heptahydrate | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Zinc sulfate              | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|--------|----------|------------------------------|---------------------------|--------------------------------------------|
|           |        |          |                              |                           |                                            |

|                           |           |                |                |                |                |
|---------------------------|-----------|----------------|----------------|----------------|----------------|
| Zinc sulfate heptahydrate | 7446-20-0 | Not applicable | Not applicable | Not applicable | Not applicable |
| Zinc sulfate              | 7733-02-0 | Listed         | Not applicable | Not applicable | Not applicable |

| Component                 | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Zinc sulfate heptahydrate | 7446-20-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y23                      |
| Zinc sulfate              | 7733-02-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y23                      |

## 16. Other information

**Prepared By**

Product Safety Department  
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www.thermofisher.com

**Creation Date**

02-February-2010

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25-March-2024

**Print Date**

25-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**